

ACTIVITY 1: Creating the Account Class (Entity/Model Class)

Objective

Create a pure Account entity class that only manages account data using boolean returns for success/failure.

Requirements

Fields

Create a class Account with the following private fields:

Field	Type	Description
accountNumber	int	Unique account number
name	String	Account holder's full name
age	int	Account holder's age
balance	double	Current account balance
accountType	String	Type of account ("Savings" or "Current")
status	String	Account status ("Active" or "Inactive")

Constructor

Create a constructor that:

- Accepts: accountNumber, name, age, initialBalance, accountType
- Sets all fields with the provided values

- Sets status to "Active" by default

Methods

Method	Parameters	Return Type	Description
<code>deposit</code>	<code>amount</code> (double)	<code>boolean</code>	Returns <code>true</code> if successful, <code>false</code> if amount ≤ 0
<code>withdraw</code>	<code>amount</code> (double)	<code>boolean</code>	Returns <code>true</code> if successful, <code>false</code> if insufficient balance or invalid amount
<code>getAccountNumber</code>	None	<code>int</code>	Returns account number
<code>getName</code>	None	<code>String</code>	Returns account holder's name
<code>getAge</code>	None	<code>int</code>	Returns account holder's age
<code>getBalance</code>	None	<code>double</code>	Returns current balance
<code>getAccountType</code>	None	<code>String</code>	Returns account type
<code>getStatus</code>	None	<code>String</code>	Returns account status
<code>setName</code>	<code>name</code> (String)	<code>void</code>	Updates account holder's name
<code>setAge</code>	<code>age</code> (int)	<code>void</code>	Updates account holder's age

Important Notes

- **NO display methods** - pure entity class
- Methods return `boolean` to indicate success/failure
- No exceptions thrown

Task

Write the complete `Account` class in a file called `Account.java`.